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WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 6A: VHDL Types - Revisited

This Module, You Should...

- Recall when to include `numeric_std`
- Illustrate how and when to **convert** between different types
- Perform **two's complement** arithmetic using **signed/unsigned** types
- Recall the seven groups of operators and their **precedence**
- Explain how **packages** work and how to construct your own package
- Demonstrate how to perform shift and rotate operations on signed/unsigned types
- Construct a VGA controller to meet the requirements for Lab 4

Type Conversions

- Why is it necessary to understand type conversions
 - Some examples
- Need to understand the different types to understand how to convert between types
- Note: Some of the material presented in the next few slides was extracted from:
 - Jim Lewis, “VHDL Math Tricks of the Trade”, MAPLD, 2003

Common VHDL Types 1

Type	Value	Origin
std_ulogic	'U', 'X', '0', '1', 'Z', 'W', 'L', 'H', '-'	std_logic_1164
std_ulogic_vector	array of std_ulogic	std_logic_1164
std_logic	Resolved std_ulogic	std_logic_1164
std_logic_vector	array of std_logic	std_logic_1164
unsigned	array of std_logic →	numeric_std, std_logic_arith
signed	array of std_logic →	numeric_std, std_logic_arith
boolean	true or false	standard
integer	$[-2^{31} - 1, 2^{31} - 1]$	standard
time	1fs to 1 hr	standard

Common VHDL Types 2

- **Resolved:**

- Function in the package that decides what the final outcome of a signal is when two or more drivers drive the same signal

- Assume the example:

```
{ a <= '1';  
  a <= '1' or '0';
```

- If `a` is `std_logic`, compiler does not complain
 - `std_logic` allows multiple drivers and will **resolve** the value
 - In this case, `a` gets '1'
- If `a` is `std_ulogic`, compiler will complain

Common VHDL Types 3

- Assume for example

$\left\{ \begin{array}{l} a \leq '1'; \\ a \leq '1' \text{ and } '0'; \end{array} \right.$

- Signal `a` will evaluate to 'X' if `std_logic` (contention/conflict)
- Compiler will complain if `a` is `std_ulogic`